Detecting Deepfakes with Metric Learning

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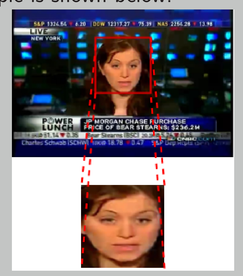
Abstract

- We investigate the effect of metric learning for deepfakes classification in high compression factor. We investigated the triplet loss against some traditional approaches to classify videos as fake or real.
- In the process, we prove that earlier approaches are heavily dependent upon the quality of videos. We also observe that these approaches fail to classify videos in high compression factor.
- We conclude that our method is more robust across different compression factors. Our work using semi-hard online triplet mining achieves AUC score of 99.7% in case of Celeb-DF (Raw videos) and 93% in case of NeuralTexture (Low Quality(LQ) videos).

Experimental Analysis of Established frameworks

- We first analyze the performance of some established deep learning approaches, which are successful elsewhere, for the task of deepfake detection.
- Using **MTCNN**, we extract faces from frames.MTCNN works well in hazy frames extracted from high compressed videos. An example is shown below:



- **2D-CNN:** In this, we use Xception architecture to classify faces as fake or real. It was able to capture dissimilarities in raw videos, however, the performance drops significantly for low quality videos.
- **Sequence Classification:** Dense layer features of 16/32 frames per videos of 512 dimension were extracted from trained Xception Net weights. Then, the network was trained with a mix of LSTM/GRU layers. It didn't help us in our case to capture the distribution of sequence in temporal domain.
- **3D Convolution:** 3D spatio-temporal filters learns the features across spatial as well as temporal dimension simultaneously. We took 32 frames per video, however, with no prior knowledge like models pre-trained on imagenet dataset, the performance of 3D networks was very poor.

Triplet Network

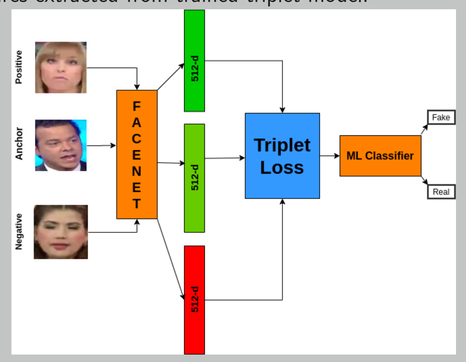
- The triplet loss is typically used in conjunction with a Siamese network which enables a metric learning framework where similar features are clustered together and different features are placed large apart in feature space.
- Let's take an anchor sample A, a sample with same label as P, and a sample with a different label N. The network tries to minimize the distance between A & P and at the same time maximizes the distance between A & N. Mathematically speaking, the loss function is defined as:

$$L(A,P,N)=max(\|f(A)-f(P)\|^2-\|f(A)-f(N)\|^2+\alpha,0)$$

- In semi-hard triplet mining, the training data is chosen such that the distance between A & N is between the distance between A & P and distance between A & P plus margin α i.e., $d(a,p) < d(a,n) < d(a,p) + margin$. The loss propagated backwards is positive and zero in this scenario.

Training and Testing

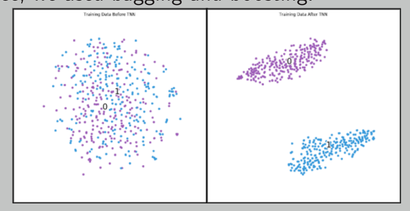
- **Training:** MTCNN extract faces from frames, then, facenet generates 512 dimension embeddings for each face feature space. Each unique face occupies a small cluster. We generate semi-hard triplets using these embeddings and apply triplet loss to group the embeddings of original and manipulated faces separately. After that, we used ML classifiers to classify the video as fake or real using the features extracted from trained triplet model.



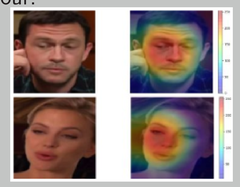
- **Testing:** We took 25 frames per video and then generate a feature vector using pre-trained triplet network. Using the learned ML classifiers weights on the training data, we classify the feature vector of the test video as fake or real.

Celeb-DF Analysis

- Train/Test Distribution: 6.5k videos were distributed into 6k for training and 500 for testing. In training, we had around 5.2k fake videos and 700 and real videos. To counter data imbalance, we used bagging and boosting.



- We employed Gradient-weighted Class Activation mapping to highlight the important regions in the image to visualize the model behaviour.



- Our approach achieves an AUC score of 99.2% with less number of frames used at the time of training outperforming other models.

FaceForensics++ Analysis

- Train/Test distribution: 1.7k training videos and 300 videos for testing with 1:1 distribution. We analyzed the behaviour of several approaches on low quality videos. Triplet networks provides a sharp gain over traditional approaches with less number of frames.

Method	Accuracy
2D-CNN	49.82
3D-CNN	43.47
LSTM/GRU	55.8
Triplets	86.74

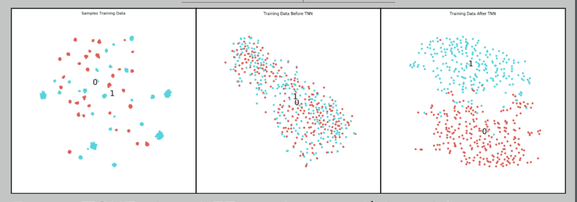
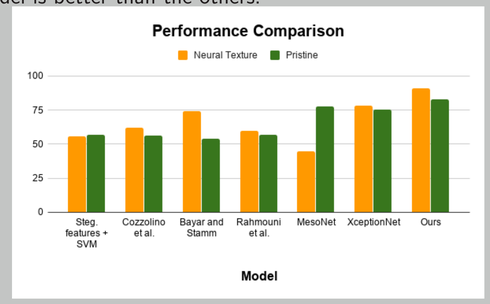


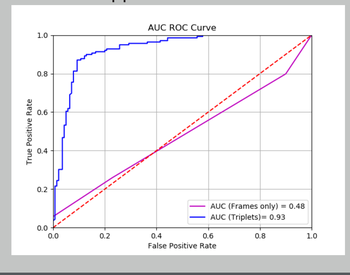
Figure: TSNE plot of FF++ dataset: 1) Initial face embeddings; 2) Embeddings before training; 3) Embeddings after training using Triplet loss

FaceForensics++ Results

- We compared the classification performance on Neural Texture videos and Pristine videos. In both the cases, our model is better than the others.



- ML classifiers on top of triplet network improves the AUC score to 93% and Equal Error Rate below 10%, that shows the robustness of our approach.



Conclusion

- We demonstrated the effectiveness of a triplet loss based deep learning approach on binary classification of deepfakes.
- Improved the AUC score by 33% on Celeb-DF (raw) dataset and 12% accuracy in FaceForensics++ (LQ) dataset.
- Incorporated less data for training and requires only 1 sec of video at the time of testing to classify video as fake/real.

References

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